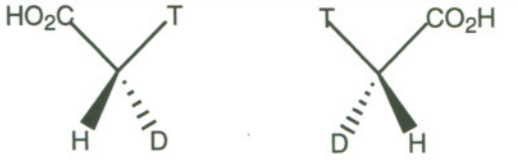
Chiral acetic acid derivatives [(*R*)-**1** and (*S*)-**1**, Figure 1] have been invaluable for mechanistic determinations in a variety of biochemical transformations. A paper in the last issue of *J. Am. Chem. Soc.* (Dehnhardt, C., *et. al.* **1999**, *121*, 10848-10849) reported an efficient approach to the synthesis of this chiral tritium, deuterium, hydrogen-substituted methyl derivative.

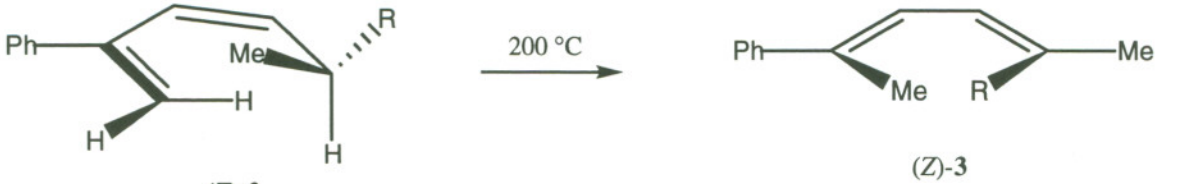


(*S*)-**1** (*R*)-**1**

Figure 1

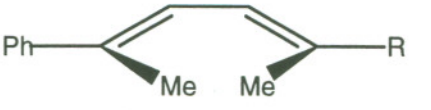
The authors chose a route to chiral **1** through a stereocontrolled reaction which would produce a T,D,H-substituted chiral methyl group for oxidative degradation to the desired (*S*)-**1**. In order to optimize stereocontrol in the reaction, they examined substiuent effects in the following reaction (Scheme 1):

Scheme 1



(*Z*)-**2** (*Z*)-**3**

+



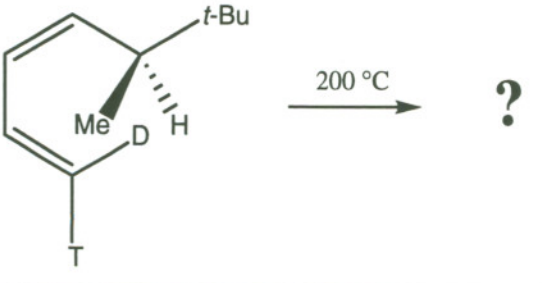
(*E*)-**3**

Derivative	R	(*E*)-**3** / (*Z*)-**3**
a	Et	1.5 : 1.0
b	*i*-Pr	2.5 : 1.0
c	*c*-hex	3.2 : 1.0
d	*t*-Bu	>100 : 1

1) What is the reaction in Scheme 1 called?

2) Why does the (*E*)/(*Z*) ratio shift to stereospecific formation of the (*E*)-isomer as the R group is changed from Et to *t*-Bu? [Explain by showing the conformational isomers of (*Z*)-**2** which lead to (*Z*)-**3** and (*E*)-**3**]

3) Scheme 2 shows the starting material used to produce the desired chiral methyl for subsequent conversion to (*S*)-**1**. Show the reaction mechanism and stereochemistry of the resulting chiral product in Scheme 2.

4) Describe the predicted stereocontrol in this reaction according to the classic rules which have been derived to describe these types of reactions.

Scheme 2